

1 Open Diagnosis and the Autonomous Mastery Runtime

This development stage removed two important forms of development-authored scaffolding from AION’s canonical cognitive runtime. First, a failed action no longer requires an evaluator to supply a clean label identifying whether the cause was a false assumption, a poor decision, execution failure, misleading evidence, environmental change, or missing authority. Second, a broad learning mission no longer needs to be expressed using the retained five-specialist vocabulary or accompanied by a complete capability graph.

Two procedures were internally promoted:

- `procedure_open_diagnostic_experiment_invention_v1`; and
- `procedure_autonomous_mastery_runtime_v1`.

Together they implement the persistent chain

$$\begin{aligned} \text{owner mission} &\rightarrow \text{learning-target induction} \rightarrow \text{specialist invention} \\ &\rightarrow \text{dependency graph} \rightarrow \text{weakest ready capability} \rightarrow \text{curriculum} \\ &\rightarrow \text{pre-action criticism} \rightarrow \text{committed action} \rightarrow \text{outcome} \\ &\rightarrow \text{open diagnosis} \rightarrow \text{reflection or private repair} \rightarrow \text{transfer test} \\ &\rightarrow \text{retention} \rightarrow \text{persistent mastery update.} \end{aligned} \tag{1}$$

This extends the existing canonical runtime rather than introducing a competing goal or cognition loop. HexCore continues to govern action, independently revealed consequences determine success, and the Cognitive Authority Unit (CAU) remains the promotion authority.

1.1 Self-Generated Pre-Action Criticism

Earlier metacognitive review could evaluate counterfactuals and alternative actions supplied by a planner. The new controller also detects when a consequential action contains no meaningful opposition. For each material assumption a_i , it constructs the unverified counterfactual

$$c_i : \quad \neg a_i, \tag{2}$$

and asks which observable consequence would distinguish a_i from $\neg a_i$. If no explicit assumption exists, it constructs the generic opposition that the action may execute correctly while failing to achieve the intended outcome. Each resulting information action is marked proposal-only, reversible, low risk, and dependent on an independent outcome.

The generated criticism is not accepted as truth. Its purpose is to expose an untested assumption or missing discriminator before consequential commitment. The mechanism was evaluated on software, research, causal, physical, and planning decisions. All five generated independent counterfactual questions and information actions without a language-model call.

1.2 Open Diagnostic Experiment Invention

When an observed failure has several plausible causes, AION retains a live hypothesis set

$$\mathcal{H} = \{h_1, \dots, h_n\} \tag{3}$$

Table 1: Self-generated pre-action criticism results.

Measure	Result
Consequential decision families	5
Counterfactual sets generated	5/5
Information-action sets generated	5/5
Actions granted independent authority	0
LLM calls	0
Unsafe executions	0

rather than forcing an immediate label. Each safe observation contract q contains a read-only operation, an independent authority, an action cost $C(q)$, and the observation predicted by every h_i . The induced partition of the live hypotheses has entropy

$$I(q) = - \sum_{v \in V_q} p(v) \log_2 p(v), \quad (4)$$

where V_q is the set of distinct predicted observations. AION selects

$$q^* = \arg \max_{q \in \mathcal{Q}_{\text{safe}}} \frac{I(q)}{C(q)}. \quad (5)$$

The selected experiment and hypothesis set are SHA-256 committed before the outcome authority reveals a result. Hypotheses inconsistent with that result are removed and the process repeats. If no safe informative observation remains while more than one hypothesis survives, AION abstains.

Six sealed cases covered assumption, decision-model, execution, evidence, environment, and authority failures. Surface channel names were changed in every case, preventing memorisation of one fixed diagnostic sequence.

Table 2: Open diagnostic experiment-invention results.

Measure	Result
Ambiguous failure cases	6
Correctly resolved origins	6/6
Diagnostic accuracy	100%
Mean diagnostic actions	2.667
Unsafe diagnostic actions	0
Forced diagnoses without sufficient evidence	0

The mechanism therefore changes post-action reflection from passive labelling into active causal discrimination:

$$\text{poor outcome} \rightarrow \text{competing causes} \rightarrow \text{diagnostic intervention} \rightarrow \text{evidence-bound lesson}. \quad (6)$$

1.3 Open Learning-Target and Specialist-Contract Invention

The mastery compiler receives a broad owner objective, permitted outcome authorities, and an action budget. It identifies explicit learning targets but does not infer a terminal purpose from

arbitrary nouns. If the objective does not state what is to be learned or mastered, it requests clarification. If no independent outcome authority is available, it returns a needs-authority state.

For every resolved target t , AION invents target-specific contracts for four generic mastery functions:

$$E_t \prec P_t \prec T_t \prec R_t, \quad (7)$$

where E_t denotes evidence-model construction, P_t deliberate practice, T_t source-disjoint transfer, and R_t delayed retention. Each specialist contract declares:

$$S_t = (I, O, D, A, V), \quad (8)$$

where I and O are its typed inputs and outputs, D is its prerequisite set, A is the accepted outcome authority, and V is the verification contract. Every contract remains proposal-only.

For objectives containing several targets, an additional integration contract depends on all target-level retention nodes. A topological check rejects a cyclic or unresolved graph before mission authorization.

The sealed mission contained four targets:

1. Rust systems engineering;
2. accessible user-interface design;
3. evidence-grounded financial-market analysis; and
4. bioacoustic event classification.

The fourth target was intentionally outside the retained five-domain specialist vocabulary. The compiler produced four contracts per target plus one cross-target integration contract, giving 17 invented specialists.

1.4 Persistent Weakness-Driven Mastery Control

For mission M and capability c , AION retains attempts n_c , verified successes s_c , transfer successes t_c , mean score $\bar{x}_c$, completed task identifiers, and the Wilson lower-confidence estimate

$$\ell_c = \frac{\hat{p}_c + \frac{z^2}{2n_c} - z\sqrt{\frac{\hat{p}_c(1-\hat{p}_c)}{n_c} + \frac{z^2}{4n_c^2}}}{1 + \frac{z^2}{n_c}}, \quad \hat{p}_c = \frac{s_c}{n_c}, \quad z = 1.96. \quad (9)$$

Only dependency-ready, unmastered capabilities are eligible. The next target maximises strategic weakness,

$$c^* = \arg \max_{c \in \mathcal{U}(M)} w_c(1 - \ell_c), \quad (10)$$

where w_c gives additional value to transfer and integration contracts.

Candidate learning methods are compositions of generic epistemic operators rather than per-domain answer strings. Only independently verified outcomes update method ordering. A successful method in one target can therefore reduce the number of evaluations required for a later target, while the later target must still pass its own authority and transfer contract.

An initial restart audit exposed a reporting defect: a completed mission correctly generated no new tasks, but the summary divided by an empty list of new attempts. The state machine had preserved the right cognition; the audit layer had not handled a zero-work restart. The calculation was replaced with the retained outcome receipts, and a regression test now verifies that a completed mission retains exactly 34 outcomes and creates no duplicate work.

Table 3: Autonomous Mastery Runtime sealed results.

Measure	Result
Owner learning targets induced	4
Specialist contracts invented	17
Capability contracts completed	17/17
Independently verified outcomes	34/34
Attempt reduction versus cold composition	56.86%
Terminal-objective mutations	0
Unsafe live writes	0
Restart retention	100%
Duplicate tasks after completed-mission restart	0
Focused integrated regression tests	29/29

1.5 Canonical Runtime Integration and Authority Boundary

The canonical runtime now exposes `compile_and_authorize_mastery_mission`. Compilation and authorization are deliberately separate operations. By default, the compiler returns an invented graph as a proposal. An authorized caller must explicitly set the authorization boundary, after which the terminal objective hash is immutable:

$$H_M = \text{SHA256}(\text{terminal objective}), \quad H_M^{(t+1)} = H_M^{(t)}. \quad (11)$$

AION may invent instrumental goals, curricula, tools, diagnostic experiments, specialists, and dependency structures serving H_M . It cannot silently replace H_M or authorize its own terminal mission.

Post-action reflection is also connected to open diagnosis. When a critic reports ambiguity and supplies competing predictive models plus safe observation contracts, the canonical runtime retains the proposed diagnostic experiment. When no informative safe contract exists, the result remains an abstention rather than becoming a guessed lesson.

At completion of this cycle, the canonical cognitive runtime, continuous real-outcome learner, and Arena v15 long-duration campaign remained active. The verified trajectory manifest contained 139 committed artifacts, including 106 verified positives and 21 verified negatives, with zero procedure-split leakage.

1.6 Claim Boundary and Real-Apprenticeship Gate

This result establishes a persistent semi-autonomous mastery *control architecture*. It does not establish that AION has already mastered Rust, interface design, finance, or bioacoustics. The generic four-phase mastery scaffold, sealed method-property authority, diagnostic prediction contracts, and learning-target grammar remain engineered.

The next decisive stage is real apprenticeship. An owner-authorized target must be connected to authoritative resources, actual compilers or applications, unfamiliar projects, and independently revealed execution, user, market-data, or sensor consequences. A credible mastery record must then demonstrate:

1. curriculum and experiment generation without a supplied exercise list;
2. repeated successful work rather than one demonstration;

3. source-disjoint transfer to unfamiliar projects;
4. delayed retention without rereading or outcome replay;
5. correct active diagnosis of naturally occurring failures;
6. no serious regression of protected capabilities; and
7. explicit abstention or owner escalation outside delegated authority.

Only those real consequences can change the claim from “AION can organise and govern autonomous mastery” to “AION acquired deep transferable competence in a real subject.”